OPN (MAIN PLANT) Department
SIMHADRI

Note No :1

Ref:SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917 **Date:**24/11/2024(17:00:11)

Subject:UNIT PROTECTION DEVICES

LMI/OD/OPS/SYST/007 **REV-08 DATE- Nov 2024 "UNIT PROTECTION DEVICES"** has been reviewed as Periodical Review. **No changes made in this revision.**

Put up for kind Approval.

Submitted by:

जी शेखर - वरिष्ठ प्रबंधक

G Sekhar - 097659 (SENIOR MANAGER ,OPN (MAIN PLANT)) ,Simhadri (GSEKHAR@NTPC.CO.IN, 8331015818)
On:24/11/2024(17:00:11) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:2

Reviewed and is being forwaded for further approval by competent authority.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri
On:25/11/2024(15:38:30) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:3

Checked and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri
On:01/12/2024(14:24:12) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:4

Please review.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG) (NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:02/12/2024(10:57:28) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:5

LMI reviewed

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG) (AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:04/12/2024(11:23:42) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:6

LMI attached is in line with the OD.

May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:21/12/2024(15:41:31) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:7

The subject LMI has been reviewed and is forwarded for further review.

Submitted by:

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

Suresh Babu Kolla - 007980 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(SBKOLLA@NTPC.CO.IN, 9490307556) ,Simhadri
On:23/12/2024(10:11:59) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:8

Please review the LMI

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE) (NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri
On:23/12/2024(19:58:47) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:9

As discussed please review.

Submitted by:

के पी रमेश - वरिष्ठ प्रबंधक

KANADE PIYUSH RAMESH - 101897 (SENIOR MANAGER ,MECH MAINT- TURBINE) (PIYUSHKANADE@NTPC.CO.IN, 9425219272) ,Simhadri
On:23/12/2024(22:22:24) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:10

Reviewd and necessory changes made.

Submitted by:

जी शेखर - वरिष्ठ प्रबंधक

G Sekhar - 097659 (SENIOR MANAGER ,OPN (MAIN PLANT))
(GSEKHAR@NTPC.CO.IN, 8331015818) ,Simhadri
On:24/12/2024(21:08:11) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:11

LMI as per attached annexure titled "Unit Protection Devices edited" is reviewed and forwarded for approval.

Submitted by:

के पी रमेश - वरिष्ठ प्रबंधक

KANADE PIYUSH RAMESH - 101897 (SENIOR MANAGER ,MECH MAINT- TURBINE)
(PIYUSHKANADE@NTPC.CO.IN, 9425219272) ,Simhadri
On:25/12/2024(01:07:07) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:12

Forwarded for approval

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri
On:26/12/2024(09:29:08) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:13

May pl review the LMI document for further review

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri
On:26/12/2024(18:37:07) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:14

Submitted by:

सुनेता साहा - उप महाप्रबंधक

Suneta Saha - 006843 (DY. GENERAL MANAGER ,C & I MAINT)
(SUNEITASAHA@NTPC.CO.IN, 9434737713) ,Simhadri
On:26/12/2024(19:25:14) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:15

LMI Unit Protection Devices Edited reviewed.

forwarded for further approval.

Submitted by:

प्रदीप कुमार मौर्य - वरिष्ठ प्रबंधक

PRADEEP KUMAR MOURYA - 100966 (SENIOR MANAGER ,C & I MAINT)
(PRADEEPMOURYA@NTPC.CO.IN, 9650995380) ,Simhadri
On:29/12/2024(11:35:21) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:16

Submitted by:

सुनेता साहा - उप महाप्रबंधक

Suneta Saha - 006843 (DY. GENERAL MANAGER ,C & I MAINT)
(SUNEITASAHA@NTPC.CO.IN, 9434737713) ,Simhadri
On:30/12/2024(16:32:03) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:17

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri
On:01/01/2025(21:00:28) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:18

Please review.

Submitted by:

भास्कर वी के कुरीचेट्टी - उप महाप्रबंधक

Bhaskar V K Kurichety - 008045 (DY. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri
On:02/01/2025(10:54:32) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:19

Reviewed and forwarded for approval

Submitted by:

चौ. श्रीवेंकट फणी कुमार - वरिष्ठ प्रबंधक

SRIVENKATA PHANI KUMAR CH - 100247 (SENIOR MANAGER ,MTP)
(CHSRIVENKATAPHANIKUMAR@NTPC.CO.IN, 9440100534) ,Simhadri
On:03/01/2025(15:36:33) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:20

Reviewed and forwarded for approval.

Submitted by:

भास्कर वी के कुरीचेट्टी - उप महाप्रबंधक

Bhaskar V K Kurichety - 008045 (DY. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri
On:03/01/2025(15:41:28) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:21

LMI attached is in line with the OD.
May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:06/01/2025(10:19:10) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:22

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri
On:10/01/2025(09:37:05) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

Note No:23

Approved & Submitted by:

समीर शर्मा - मुख्य महाप्रबंधक

Sameer Sharma - 005041 (CGM ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri
On:10/01/2025(10:45:54) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPR_LMI/699917

NTPC SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

UNIT PROTECTION DEVICES

LMI/OD/OPS/SYST/007 REV:08

NOVEMBER-2024

Approved for Implementation by HOP (SIMHADRI)

TITLE	UNIT PROTECTION DEVICES
LMI No.	LMI/OD/OPS/SYST/007 REV:08

LMI APPROVAL DOCUMENT

1	**LMI NAME**	UNIT PROTECTION DEVICES
2	**LMI No.**	LMI/OD/OPS/SYST/007 REV:08 NOVEMBER-2024
3	**Reference Document**	COS-ISO-OD/OPS/SYST/007 SEPTEMBER-2022
4	**Superseded Document**	LMI/OD/OPS/SYST/007 REV:08 NOVEMBER-2022
5	**Prepared By**	G Sekhar Senior Manager (Operation)
6	**Checked By**	Malay Kanti Jana AGM (Operation)
7	**Checked By**	Somnath Bhattacharya AGM (Operation)
8	**Checked By**	Navin Kishore AGM (EEMG)
9	**Checked By**	Suresh Babu Kolla AGM (BMD)
10	**Checked By**	Nilanjal Mandal AGM (TMD)
11	**Checked By**	Jayamol Rachel Varughese AGM (C&I)
12	**Checked By**	Bhaskar V K Kurichety DGM (MTP)
13	**Rechecked & Reviewed By**	Prasanta Kumar Jena GM (O&M)
14	**Approved By**	Sameer Sharma CGM (Simhadri)

TITLE	UNIT PROTECTION DEVICES
LMI No.	LMI/OD/OPS/SYST/007 REV:08

CONTENTS

TITLE	UNIT PROTECTION DEVICES
LMI No.	LMI/OD/OPS/SYST/007 REV:08

1. Introduction

Plant protection encompasses various interlocks and protection systems. The term interlock predetermines the sequential order in which certain operations may be initiated to commence a start up and /or initiate automatic trip out in the event of unsafe process condition or initiation of unsafe operation. The prevention of action by an interlock or the tripping of plant is accompanied by displaying to the operator this action has been taken. There is combination with alarm displays and predetermined set of operational instructions to minimize the risk of damage to the equipment and provide safeguards to equipment and personnel.

In order that the interlocks and protections of equipment perform required duty, when they are called upon, it is essential to ensure highest degree of integrity, reliability and availability.

1.1. Interlock and protection devices in a power station can be broadly classified as:

1.1.1. An interlock is a feature that makes the state of two or more mechanisms or functions mutually dependent. It may be used to prevent undesired states in a finite-state machine, and may consist of any electrical, electronic, or mechanical devices or systems Interlocks which prevent plant operation until the nominated permissive conditions are satisfied and in turn safeguard the Boiler, Turbine and Generator against process parameter deviations and are termed as UNIT PROTECTION.

1.1.2. Main Unit Overall Protection- This is a protection which executes Turbine protection and Unit electrical protection and provides inter tripping to the boiler protection equipment.

1.1.3. Any protection which trips running equipment to avoid direct damage to the equipment/plant is termed EQUIPMENT PROTECTION.

1.1.4. Safety devices which acts to release excessive energy at preset value (e.g. safety and relief valves), thereby safeguarding the plant and personnel against sudden escape of energy. These are identified as "SAFETY /RELIEF DEVICES".

1.1.5. Electrical Trip Protection provided to safeguard electrical devices against overload short circuit and any other type of electrical system faults, are identified as ELECTRICAL PROTECTION.

2. Superseded Documents & Revision Notes

- Superseded: LMI/OD/OPS/SYST/007, Rev No: 2 dated: Sept 2014 Rev01 June'2010: Boiler Protection logics modified.
- Rev02 Sept'2014: Stage#2 Unit 3&4 scope included. Rev03 Dec'2017: Revised in line with OD revision.
- Rev04 Jan'2020: Revision notes introduced in paragraph 2.0. Rev05 May 2020:
    a) In paragraph no 5.0 Frequency of Protection checking added.

b) In paragraph no 6.1, 6.2.1 & 6.2.2 presence of EEMG during joint checking of protections has been included.

c) Standard Protection checking Format is attached as annexure I.

- Rev06 June' 2022: LMI has been revised as per periodical review of 2 years period. No modification made in this revision.
- Rev 07 Nov 2022 : LMI has been revised as per COS-ISO-00-OD/OPS/SYST/007 Rev-03 Sept 2022
- REV 08 NOV 2024: LMI format changed as per new guidelines

Supersedes and merges

a) LMI-OD-OPS-SYST-006 Issue-1, Rev 03 Jan-2022

b) LMI-OD-OPS-SYST-007 Rev-6, June-2022

c) LMI-OGN-OPS-MECH-003 Rev-02, Mar 2022

3. Scope

I. This LMI is applicable for checking of "UNIT PROTECTION" "Main unit overall protection " as described in item 1.1.1, 1.1.2 and equipment protections as described in item 1.1.3 in 4 X 500 MW Units of Simhadri STPS, Stage-I & II

II. Item 1.1.5 is addressed in LMI "Testing of electrical protections" OD/OPS/Elec/003 and the records are maintained by Electrical Maintenance.

III. Item 1.1.5 is not covered in the scope of this LMI.

4. References

This LMI is based on Operation Directive OD/OPS/SYST/007, Rev No.: 3 Date : Sept. 2022.

5. Frequency Of Protection Checking:

5.1 Permissive and Protection checking of Boiler, Turbine, and Generator are to be carried out as follows:

5.1.1 Checking of protection, permissive & interlocks during Re-commissioning after unit overhauls.

5.1.2.1 During an outage of more than 48 hours occurring after 3 months of last such check, the Inter trips associated with Boiler, Turbine and Generator shall be checked by Operation along with C&I, EMD, EEMG and concerned maintenance group. Signed protocols shall be put up for the approval of Head of (O&M) before the starting of the unit.

5.1.3 During outages extending more than 48 hrs. Occurring three months after the last such checks, all protections relating to the Boiler, Turbine & generator shall be checked by Operation along with C&I, EEMG and concerned maintenance group. Signed protocols shall be put up for the approval of Head of (O&M) before the starting of the unit.

5.1.4 If any major work is carried out on any equipment or any element or part of any element used for protection of the equipment or unit protection is replaced, removed or recalibrated then protection & interlock of the equipment to be checked before taking the equipment in service .

5.2 Permissive and Protection of major unit auxiliary equipment such as BFPs, CEPs, LP/HP Heaters, CW Pumps, I.D. /F.D /P.A. fans, Mills, Generator cooling and sealing system equipment, Air preheaters, SGECW, TGECW, ACW pumps etc. are to be checked during recommissioning after overhaul / major work.

5.3 Permissive and Protection of All other auxiliaries (including offsite and CHP): Such as IAC, PAC,TAC, Fire water hydrant and spray pumps, Raw water pumps, DG sets, Ash Raw water pump, Make up water pumps, Ash handling Pumps, Ash water pumps, Air conditioning equipment, crushers, conveyor belts, tripper, paddle feeders, equipment of FGD units etc. are to be checked during recommissioning of the equipment after overhaul / major work or otherwise annually.

6. Mode & Methodology Of Checking

6.1 All Protections & Interlocks to be checked during Re-commissioning unit overhauls:

6.2 All protection systems and interlocks shall be checked jointly by Operation, C&I maintenance, EEMG and the respective Maintenance Department. The deviations from the envisaged design shall be clearly recorded and approved by Head of O & M before the unit is recommissioned. All the sensing elements like pressure/temperature switches, relays, etc. shall be calibrated, compared with design/ previous records. Head of respective maintenance shall approve this list of Boiler, Turbine & Generator mechanical protections along with methodology to be adopted for testing is listed out in Annexure-II for Stage #1 and in Annexure-III for Stage #2.

6.3 Protocols of P&I checks to be prepared immediately after completion of checks duly signed by all concerned.

6.4 Wherever it is possible to create actual conditions, without endangering the safety of plant and/or personnel, the protection shall be checked by varying the required parameter/ injecting signal.

6.5 Wherever it is not possible, checking shall be done by simulating the condition at the sensing field element.

6.6 Any signal /condition/input forced or simulated during the checking of the protection/permissive/Interlock, should be normalized immediately after completion of checks & same to be recorded in Protocols.

6.7 If any unit protection is unhealthy, the same to be rectified before unit light up or If unit is in running condition OD on protection bypass to be followed.

7. Turbine Protection Devices

a) Healthiness of turbine protection device to be checked in regular intervals when the machine is on load/or off load regularly to ensure reliability and to highlight problems & initiating necessary corrective action.

b) Turbine protections devices are to be tested when a machine is on load. Testing to be carried out using Automatic Turbine Tester (ATT) or equivalent system if provided.

c) TG protection devices /equipment/ of different manufacturers have recommended varying frequency of testing. All locations required to follow the frequency of testing as per manufacturers recommendations applicable to the machines in their stations. If no manufacturers testing frequency is specified, then testing shall be carried out at an interval as indicated in table no.2

7.1. Main Turbine Protection testing Requirement

Following table contains details of all test with frequency of test .

SN	Equipment	Purpose	Mode of Testing	Frequency
7.1.1	Turbine stop valves	a) Smooth operation and to avoid carbonization of the spindle at the gland area. b) To check the time necessary for closure c)To ensure smooth & jerk free operation over the full range of travel	Through Automatic Turbine trip Testing (ATT) Wherever provided. (in Remote mode only)	As per OEM Recommendation. In the absence of OEM Recommendation, testing shall be carried out at an interval not exceeding three months or any available opportunity.
7.1.2	Turbine control valves	To ensure smooth of operation and to avoid carbonization of the spindle at the gland area. To ensure smooth & jerk free Operation over the full range of travel	ATT or (In Remote mode only)	
7.1.3	Overspeed trip device by oil Injection test	To prevent the trip bolts from jamming. To predict whether these bolts will act at the rated overspeed values/ Test oil pressure.	ATT or (In Remote mode only) Or manually from Gov. rack	The testing shall be carried out at an interval not exceeding one month.
7.1.4	Low Vacuum trip device	To ensure operation of the Mechanical device at rated conditions.	ATT (In Remote mode only)	As per OEM Recommendation. In the absence of OEM Recommendation, testing shall be carried out at an interval not exceeding ONE months or any available opportunity / during startup if unit

				shutdown greater than one month .
7.1.5	Remote Trip Solenoid	To ensure satisfactory operation	ATT (In Remote mode only)	As per OEM Recommendation. In the absence of OEM Recommendation testing shall be carried out at an interval not exceeding one months or any available opportunity / during startup if unit shutdown greater than one month .
7.1.6	Main trip valve	To ensure satisfactory operation	ATT (In Remote mode only)	During every unit overhaul or any available opportunity. During every unit Overhaul, trip oil pressure at which, main trip valve(MTV) operates(trips) to be recorded by closing the CF isolation valve provided in protection rack at EL 17M floor.
7.1.7	Extraction Block valve	Part operation to ensure that they free to close when required.	Remote /Manual	Once in 6 months or any available opportunity.

7.1.8	Extraction NRV's	Part operation to ensure that they free to close when required	Remote /Manual	Once in 6 months or any available opportunity.
7.1.9	DC Lube Oil Pumps	To ensure that the Pump operates when required and develops satisfactory oil pressure.	Remote & once in a week from local PB	Daily basis
7.1.10	Heater drip level protection and alarms.	To ensure its correct operation.	Remote	During every unit overhaul
7.1.11	Deaerator over-flow protection and alarms	To ensure its correct operation.	Remote /Manual	During every unit overhaul
7.1.12	Extraction drains valves	To ensure satisfactory operation.	Remote /Manual	Once in a month
7.1.13	Turbine on line vibration instruments	To ensure satisfactory operation.	Check by using offline vibration monitoring equipment's	Once in a month

7.1.14	Turbovisory Instruments	To ensure satisfactory operation of the indicating/ recording instruments.	Remote /Manual	During every unit overhaul or whenever TSI values cross alarm values.

7.2	**Action Plan for deviations observed during checking of protection device**		
	Equipment	**Deviations observed**	**Action Plan**
7.2.1	Turbine stop valves	If any stop valve failed to operate or not operating smoothly or not closing fully	SOP to be prepared before Unit is withdrawn to avoid any eventuality.
7.2.2	Turbine control valves	If any control valve fail to operate or having jerky operation.	SOP to be prepared & Unit load to be reduced to tech minimum & pilot valve cleaning to be done
7.2.3	Main trip valve	If main trip valve failed to operate or operating at lower or higher trip oil press.	If main trip gear is failed to operate in ATT then SOP to be prepared to withdraw the unit under such condition & unit to be withdrawn. if it is operating at rated press ,same need to be attended during shutdown or overhaul.
7.2.4	Overspeed trip device	If any overspeed trip bolt fail to operate or operating at higher press	Unit to be Withdrawn & problem to be rectified
7.2.5	Low Vacuum trip device	If protection is not acting or acting at delayed or poorer vac.	Unit to be withdrawn & resetting of device to be done
7.2.6	Remote Trip Solenoid	If any RTS failed to operate	SOP to be prepared before Unit is withdrawn to avoid any eventuality
7.2.7	Extraction Block valve	If any block valve failed to operate or not operating full	Respective heater to be taken out & block valve to be rectified
7.2.8	Extraction NRV's	If any NRV failed to operate or not closing full	Respective heater to be taken out & block valve to be rectified
7.2. 9	DC Lube Oil Pumps	If failed to start or not developing press or not running to desired speed	Problem to be attended on immediate priority & OD on DC failures to be referred to.

8.Protection checking of KWU turbines

This directive aims at streamlining the testing modes and procedures of various turbine testing devices of turbines, recording of test results and rectification of defects. Details related to testingof KWU turbine is attached in annexures. Checking of protection device of KWU turbine are in annexure.

9.Records to be maintained

Record Description	Maintained by	Frequency	Remarks
Records must be kept for all above tests and should contain the following data: Date of test Details of observations Exception of the above testing	Responsibility of carrying out above inspection shall be with Operation Department. However, joint protocols shall be made along with TMD and C&I Department as applicable.	As mentioned, Above.	Rectification of observed defects during available opportunity/ Unit overhaul

10. Directive

The LMI will detail the protection systems specify the actions to be undertaken by station staff, considering the design and peculiarities of the plant, methodology of testing the records to be maintained and the approving authority in each case. Station will prepare test instruction for each protection devices based on the type of equipment installed & testing philosophy adopted by OEM.

11 Annexures

11.1 Test Instruction- Over speed trip device

S.No	DESCRIPTION
1	**Test mode**: The test must be performed at normal rated speed if the turbine is not running at rated speed, the trip initiating pressures are altered

2	• Procedure • Depress the left hand spindle of the overspeed trip test device fully and hold it in this position until the test is completed (the valve prevents the trip activation pulse from the overspeed trips from causing an actual trip. • Before increasing test oil pressure check whether adequate aux. start up oil pressure is available for resetting after the test. For this press right hand side knob (left hand side knob is already in pressed position) and see generation of adequate aux. start up oil pressure. Afterwards release right hand side knob & proceed for injecting test oil into the system. • Turn the handwheel of the overspeed trip test device in the clockwise direction. • Observe the test oil pressure. • Read and record the trip initiation pressure. • After both the overspeed trips have operated, turn the handwheel of the overspeed trip test device in the anticlockwise direction back to the starting position. • At the same time, observe the drop in test oil pressure. • Re-latch the overspeed trip actuation devices by depressing the right-hand spindle of the test signal generator. • Observe that the auxiliary trip oil pressure has been reestablished. • If the auxiliary trip oil pressure does not increase, the relatching valve must be operated again and held in position for approx. 5 seconds. • Release the left hand spindle of the test signal generator as soon as the auxiliary trip oil pressure has been reestablished. • Before Execution of the test, the test oil pressure reading must be ensured for zero.
3	**Test result & action** • The overspeed trips operate, all alarms in the general alarm annunciation equipment have responded and no criteria alarms haveappeared on the tiles of the automatic turbine tester. • Manual testing using the overspeed trip test device has reached a satisfactory conclusion when the overspeed trips operate faultlessly atthe previously recorded test oil pressures. • Should both or any one of the overspeed trips do not operate at the previously recorded pressure even after repeated testing then an off-normal condition exits in the system which must be rectified. • In the event of spurious initiation of an overspeed trip, the test must be repeated at shorter intervals than those specified in the 'Testing Intervals' until it is proved that the overspeed trip operates faultlessly at the previously recorded pressure**.**

11.2 Test instruction: Low vacuum trip device

S.No	Description
1	**Testing method:** • By means of the automatic turbine tester, the low vacuum trip can betested while the turbine generator is running. • The low vacuum trip can be tested during turning-gear operation when vacuum is pulled in the condenser. It is done, by reducing the condenser vacuum and simulating the primary oil pressure. The primaryoil pressure is simulated by screwing in to the limit the stop screw for the primary oil piston. • This test is done during initial adjustment, during routine, precise checks of the set point, and after repairs to the equipment or overhaul of the entire plant**.**
2	**Procedure** • The following procedure is adopted for testing during turning gear operation by alteration of the condenser vacuum. • Adjust the absolute pressure in the condenser to less than 0.1 bar bymeans of the normal vacuum pump. • Close the test valves of the emergency stop valves • Operate the starting and load limit device to 40% • Unscrew the cap of the low vacuum trip stop-screw • Screw in the primary oil piston stop-screw of the low vacuum trip to thelimit • Reduce the condenser vacuum • Record the functioning and response pressure of the low vacuum trip • Observe the drop in the trip fluid pressure • Observe the response of the alarms (low vacuum trip and main tripvalve operated) • Increase the condenser vacuum • Acknowledge and cancel the alarms. • Actuate the main trip valve, using the starting and load limit device • Open the test valves • Unscrew the stop screw for the primary oil piston of the low vacuumtrip, and lock it in position • Replace the cap and screw it down so that it is oil-tight
3	**Test results** Testing with the automatic turbine tester has reached a satisfactory conclusionwhen low vacuum trip and the associated alarms in the general alarm equipment respond and no fault alarm is shown on the annunciator tile of the automatic turbine tester.
4	**Recording the tests** Protocol to be prepared with parameters & signed by Operation, C&I &TMD

11.3 Test instruction: Testing the stop valves and control valves with the automatic turbine tester

S.No	Description

1	**Test Procedure**
	Testing proceeds automatically after the automatic turbine tester has been switched on and the test program has been selected and started. For safety reason, the key switch should not be switched on until immediately before the test is to be carried out.
	The test procedure is as follows:
	• Switch on the automatic turbine tester by means of the key switch.
	• Switch on subgroup control
	• select the valve groups to be tested from the selector tile.(When several valve groups are selected they will be tested automatically in sequence).
	• Start the test programme by means of push button.
	• Observe the sequence of the test program.
	  • control valve closes
	  • stop valve closes
	  • stop valve opens
	  • control valve opens
	• Observe proper operation of the load controller which should maintain the load steady.
	• Check the annunciator tile of the automatic turbine tester for alarms.
	• When the last test program has been completed the subgroup control must be switched off.
	• Switch off the automatic turbine tester by means of key switch.
	• The status of the program can be observed at the operationtile of the automatic turbine tester by annunciations and indication lamps.
	• The signals, which appear when the program is proceeding normally, are shown.

2	**Results** At test carried out by means of the automatic turbine tester may be regarded as successful if, after the test program has been completed thetester disengages and there are no fault indications on the tester tile. **Any fault discovered during testing must be rectified as soon as Possible especially if the safety**

	of the turbine is affected

11.4 Testing the differential pressure protection of extraction valves

S.No	Description
1	**Test procedure** The test is affected by equalizing the pressure on both sides of the differential pressure switch. • Extraction valve is open, • Close shut-off valves, • Open the equalizing valve, • Observe the reduction of the pressure differential, • Observe closure of the extraction valve, • Close the equalizing valve, • Open the shut-off valves - as near as possible simultaneously.
2	**Results and action to be taken if defective** The test may be regarded as successful if the extraction valve closes as soon as the pressure differential falls below the set value. In the event of any irregularities, open the vents. Before repeating the test wait until the condensate level has built upto the condensate drain vessels

11.5 Testing the automatic starting of Auxiliary oil pumps and DC bearing oil pump

S.No	Description

Test procedure

- If the testing of the subloop control of the auxiliaryoil pumps occurs during shutdown of the turbine only one pump can be tested at a time.
- Care must be taken to ensure each of the auxiliary oilpumps are tested according to the testing recommendations.
- A testing schedule should beestablished to fulfill this requirement. Test procedure:
  - Select auxiliary oil pump to be tested according to testschedule.
  - Set the respective control switch to ensure that only the auxiliary oil pump which has beenchosen will start during test.
  - Shut-down turbine.
- Observe speed and oil pressure drop.
- Observe start of the auxiliary oil pump
- Record starting pressure and turbine speed atwhich the auxiliary oil pump started.
- If the subloop control fails to start the auxiliaryoil pump at the set oil pressure, the pump must immediately be started manually.
- During turbine standstill the testing of the subloop control of the auxiliary oil pumps and theDC bearing oil pump can be carried out as follows:

Preconditions

- Oil system depressurized
- Subloop control switched off.

Testing procedure

- Switch on subloop control.
- Observe starting of all pumps which shouldhave been started by the sub loop control.

Automatic start of the DC bearing oil pump bysubloop control can also be carried out during turbine standstill as follows:

Preconditions

- Auxiliary oil pump in operation.
- Oil system pressurized.
- Subloop control of auxiliary oil pumps switched off.
- Subloop control of the DC bearing oil pump switched on.

	Testing procedure • Shutdown auxiliary oil pump manually. • Observe start of the DC bearing oil pump. • Start auxiliary oil pump • Stop DC bearing oil pump All tests described above are successfully completedif the auxiliary oil pumps or the DC bearing oil pump are started without failure by the respective subloop controls. The turbine must not be operated with defective subloop controls for the auxiliary oil pumps and especially for the DC bearing oil pump. Any defects or deficiencies must be corrected immediately

11.6 Testing the fire protection devices

S.No	Description

1	The following operating conditions must be established before all tests are carried out.
	• Auxiliary oil pump A or B running.
	• Turbine speed greater than 70 rev/min.
	• Starting and load limiting device at a load setting at which the auxiliary starting oil pressure is reduced.
	• Main-trip gear set.
	  • Test valves of stop valves in close position.
	  • Stop valves closed.
	The following tests are carried out:
	TEST 1: Testing the functioning of the fire protection devices by operation of fireprotection switches 1 and 2.
	TEST 2: Testing the functioning of the fire protection devices by operation of fireprotection switch 2 only.
	TEST 3: Testing the functioning of the fire protection devices by operation of thepushbutton on solenoid valve 1.
	TEST 4: Testing the functioning of the fire protection devices by operation of thepushbutton on solenoid valve 2.
	The Procedure for TEST 1 is as follows:
	• **Operate fire protection switch 1**
	• Check whether all fire protection measureslisted in Fig.-1 are initiated automatically.

Component	Function initiated
Remote trip 1	On
Remote trip 2	On
Emergency shut-off valve	Close
Vacuum breaker	On

2

Check the alarm and condition annunciations listed in Fig.2 which must appear after operation of the fire protection switch 1

Component	Function initiated	Annunciation	Type of annunciation
Remote trip	On	Alarm main trip valve operated	A
Emergency shut-off valve	Closed	Desk operating control closed	C
Solenoid Valve 1	Tripped	Desk operating control closed	C
Solenoid Valve 2	Tripped	Desk operating control closed	C
Vacuum breaker	On	Desk operating control closed	C

Alarm (A) and condition (C) annunciations after operation of Fire Protection Switch 1

Operate fire protection switch 2

- Check whether all fire protection measure listed in Fig.3 are initiated automatically.

Component	Function initiated
Auxiliary oil pump 1 (SLC AOP-1)	Protection Off
Auxiliary oil pump 2 (SLC AOP-2)	Protection Off
Automatic starter for shaft lift oil pump (SLCJOP-1, JOP-2)	Off
Emergency oil pump (EOP)	On
Automatic starter for Emergency oil pump(SLC EOP)	Off
Automatic starter for turning gear (SLCTurning Gear)	Off
Automatic starter for shaft lift oil pump (SLCJOP-1)	On

Fire protection measure after operation of fire protection switch 2

12 Schedule of Turbine Protection Devices

Group	Device or test		Test requirement	Load *	Testing interval ** 1	2	3	4	5	6	7	8	Testing instruction No.:
A	Testing the overspeed trips		Rated speed	♦	•							•	I
			Overspeed					•				•	II
	Testing the main trip valve and other devices		Automatic tester	♦	•							•	III
			Oil pressure drop	♦							•	•	III
	Exercising the stop valves and control valves	Several valves with testing devices	Full travel at part load	♦								•	IV
		Several valves without testing devices	Emergency stop valves only	♦								•	IV
	Testing the stop valves and control valves with Automatic Turbine tester		Rated speed	♦	•								V
	Exercising the extraction valves			♦	•				•			•	VI
	Checking the settings of Governors and control valves		Control action as sequence of Control impulses									•	VII
	Testing the leakage of stop and control valves		Control maintains speed							•		•	VIII
			Reverse power protection				•		•			•	VIII
	Speed monitoring device			♦	•				•			•	IX
B	Automatic starters for auxiliary oil pumps								•	•	•	•	X
	Oil pressure switches											•	X
	Low-oil pressure alarms								•	•	•	•	X
C	Hydraulic low vacuum trip		Automatic tester	♦	•							•	XI
	Electr. Low vacuum trip							•	•			•	XII
D	Testing the thrust bearing trip		Automatic tester	♦	•							•	XIII
E	Differential pressure protection of extraction valves			♦	•							•	XIV
F	Fire protection devices							•			•	•	XV

a) Devices which protect from overspeed
b) Protective for oil supply
c) Overpressure protections
d) Measuring devices
e) Miscellaneous
f) Fire protection devices

*	Test under load possible ♦
**	Testing interval •
1	2 to 4 weeks
2	1 to 3 months
3	3 months to 1 year
4	1 year
5	During shut-down
6	During start-up
7	Standstill 4 to 24 h
8	After inspection

12.1 Standard Protection checking Sample Format (Boiler & Turbine)

FORMAT FOR PROTECTION CHECKING (Boiler & Turbine)

Station Name: Date:

Unit No:

Boiler/Turbine:

Document ref no:

Sl No	Description of protection	Alarm set point	Trip Set point	Time delay	Mode of protection logic *	Mode of protection checking **	Observations				Simulation Normalized	Remarks/Observations if any
							LVS Alarm (Y/N)	Alarm /Event log (Y/N)	SOE (Y/N)	Protection acted (Y/N)	(Y/N)	
1.												
2.												
3.												
4.												
5												
6												

Checked and Witnessed by

Signature &date				
Name				
Desig/Deptt	O&M/C&I	Comm. (for New unit commissioning) Operation(for commercialized unit)	C&I (Erec.)/EED(for new unit) EEMG (for commercialized unit)	Agency (for new unit)

Note: * 2 out of 3 or 2 out of 2 or 1 out of 2 etc.

** Actual/Simulation (Protections are to be checked by creating actual conditions from field. Wherever actual condition creation is not at all possible, simulation may be done for those cases only)
- In case of 2/3 logic, all combinations are to be checked.
- In case of 2/2 logic, it is to be ensured that single input does not initiate tripping.

TITLE	UNIT PROTECTION DEVICES
LMI No.	LMI/OD/OPS/SYST/007 REV:08

13.0 RESPONSIBILITY CENTER

Head of Operation is responsible for implementation.

14.0 FORMAT FOR PROTECTION CHECKING

Please refer – Format of Checking the protection at 12.1

15.0 REVIEW *: -*

This LMI is to be reviewed every Two years by the Head of Operation.

16.0 Distribution list

Simhadri Local Area Network